

Earthquake Damage Potential Calculation:

Overview

This documentation consolidates the scientific theory, data source considerations, magnitude scale differences, and recommended practical methods for calculating **earthquake damage potential based on magnitude and depth**.

1. Scientific Theory and Formula for Damage Potential Based on Magnitude and Depth

Fundamental Principles

- Magnitude (Mw)** measures the total seismic energy released by an earthquake. It is a logarithmic scale, where each unit increase means approximately 31.6 times more energy and 10 times stronger ground motion amplitude.
- Depth** affects how seismic energy attenuates before reaching the surface. Shallow earthquakes (< 70 km, especially < 30 km) cause more surface damage because energy dissipates less.
- The severity of damage correlates positively with magnitude and negatively with depth.

Empirical Damage Potential Models

Most damage assessment methodologies use **Ground Motion Prediction Equations (GMPEs)**, which model how surface shaking intensity depends on magnitude and depth. These models take the general form:

$$I = a \cdot M - b \cdot \log_{10}(d) + c$$

Where **I** is an intensity measure (e.g., Peak Ground Acceleration, Modified Mercalli Intensity), **M** is magnitude, **d** is depth, and **a, b, c** are regionally calibrated constants.

Practical Damage Potential Formula

A simplified, scientifically justified damage potential formula can be represented as:

$$\text{Damage Potential} = 0.6 \times M + 0.4 \times \frac{D_0 - D}{D_0} \times 10$$

Where:

- M = earthquake magnitude (preferably Moment Magnitude, Mw)
- D = depth (km)
- D_0 = reference maximum depth (e.g., 700 km)

This formula balances magnitude's direct effect and depth's attenuation effect on damage potential, matching empirical observations in global seismicity.

2. Data Source Considerations for Magnitude and Location

Common Magnitude Sources in Earthquake Data

- USGS (US)**: Uses multiple magnitude scales depending on event characteristics.
- ISCGEM**: Globally harmonized catalog emphasizing Moment Magnitude (Mw) or proxies.
- GCMT and HRV**: Provide rigorously calculated Moment Magnitudes using waveform models.
- ISC**: Provides authoritative location info and depth, often refined after initial reports.

Impact on Damage Potential Calculation

- Magnitudes from different sources may vary due to methodology, data availability, and saturation effects.

- Different magnitude scales (**ML**, **mb**, **Ms**, **Mw**) measure different wave types and frequency bands.
- For damage calculation, magnitude standardization to Moment Magnitude (**Mw**) is crucial to ensure comparability.

3. Magnitude Scale Types and Their Effects

Main Magnitude Types in Your Dataset

Type	Description	Saturation	Suitability for Damage Calculation
mw	Moment Magnitude (Mw variants: mwc, mww, mwb, mwr)	None	Preferred and reliable
mb	Body-wave magnitude	~6.5	Requires conversion to Mw or caution
ms	Surface-wave magnitude	~8.0	Use with saturation aware conversions
ml	Local magnitude (Richter scale)	~5.8	Limited use, unreliable at higher mags

Saturation Problem

- Saturation causes underestimation of magnitude and thus damage potential for large earthquakes.
- Moment Magnitude (**Mw**) solves this by directly representing seismic moment without saturation.

4. Recommended Practical Workflow for Feature Engineering Damage Potential

Step 1: Magnitude Type Prioritization and Standardization

- Prioritize Moment Magnitude variants (**mw**, **mwc**, **mww**, **mwb**, **mwr**) as the base magnitude.
- Convert other magnitude types (**mb**, **ms**, **ml**) to Mw using empirical conversion formulas:

```
def convert_to_mw(mag_value, mag_type):
    if mag_type.lower().startswith('mw'):
        return mag_value
    elif mag_type.lower() == 'ms':
        return 1.05 * mag_value - 0.2
    elif mag_type.lower() == 'mb':
        if mag_value > 6.5:
            return 6.5 + (mag_value - 6.5) * 1.5  # correction for saturation
        return 0.67 * mag_value + 3.2
    elif mag_type.lower() == 'ml':
        return 1.2 * mag_value - 1.0
    else:
        return np.nan
```

Step 2: Use Preferred, Harmonized Magnitude and Depth

- Use authoritative location depths from trusted locationSource agencies (ISC, USGS, etc.)
- Avoid mixing depths from less reliable sources.

Step 3: Calculate Damage Potential Using HAZUS-Style Formula

```
def calculate_damage_potential_hazus(magnitude, depth):
    actual_depth = max(abs(depth), 1.0)
    log_pga = magnitude - 3.5 * np.log10(actual_depth + 7) + 1.8
    pga = 10 ** log_pga
    damage_potential = min(10.0, max(0.0, 2.5 * np.log10(pga + 0.01) + 7.5))
    return damage_potential
```

Step 4: Quality Control and Filtering

- Prefer data with **high-quality Mw magnitudes**.
- Filter out or flag low-quality or saturated magnitude types to avoid bias in training or analysis.

5. Summary of Best Practices

Aspect	Recommendation
Magnitude Scale	Use or convert to Moment Magnitude (Mw)
Depth Source	Use consistent authoritative data (ISC, USGS)
Damage Potential Formula	Use empirically validated HAZUS/GMPE approach
Dataset Preparation	Filter or flag low-quality saturation-affected magnitudes
Documentation	Log magnitude sources and conversions for reproducibility

References

- USGS Earthquake Magnitude Types and Theory^{[1][2][3]}
- ISC-GEM Global Earthquake Catalog and Calibration Methods^{[4][5]}
- Ground Motion Prediction Equations and HAZUS Technical Manual^{[6][7][8][9]}
- Saturation Effects and Magnitude Conversion Studies^{[10][11]}